

A
PROJECT REPORT
ON
Decentralized File System

Submitted in partial fulfillment of the requirements of the degree of
“Bachelor of Engineering”

In
Information Technology

by

Parth Mishra:	26
Om Mujumdar :	30
Prathamesh Rane :	45
Kaustubh Suryawanshi:	65

Supervisor/Guide:

Asst .Prof. Seema Bhuravane



Department of Information Technology
K.C. College of Engineering and Management Studies And
Research, Thane (E)
University of Mumbai

2025- 26

CERTIFICATE

This is to certify that the project entitled “ **Decentralized File System** ” is a bonafede work of “ **Kaustubh Suryawanshi**” (65) “ **Parth Mishra**” (26), “**Om Mujumdar**” (30), “**Prathamesh Rane**”(45) submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of “**Bachelor of Engineering**” in “**Information Technology**” .

Prof. Seema Bhuravane
Supervisor



Prof. Amarja Adgaonkar
Head of Department

Dr. R.K.Pandey
Principal

Mini Project Report Approval

This project report entitled “ **Decentralized File System** by **Mr.Kaustubh Suryawanshi, Mr.Parth Mishra, Mr.Om Mujumdar, Mr.Prathamesh Rane** is approved For the degree of Bachelor of Engineering in **Information Technology** .

Examiners

1.-----

2.-----

Date:

Place:

ACKNOWLEDGEMENT

Our first foremost thanks go to my guide **Asst. Seema Bhuravne** for her guidance, help and constant encouragement throughout this project. I am thankful to those who have always been with me generously in bringing this report on “**Decentralized File System**” from the outset of the project. I am extremely obliged in expressing my thanks to **Prof. Seema Bhuravne** for their continual initiation, encouragement, and excellent guidance at every stage of the project work. I am grateful to their inspiration and valuable suggestions from time to time and providing all the lab facilities during development. I would like to thank also my friends for their support and cooperation time to time. Above all, I am thankful to all mighty god and my parents for their at most moral support and blessings without which the project would not have been success.

Thanking You.

TABLE OF CONTENT

Topic	Page No.
Abstract	ii
Acknowledgement	iii
List of abbreviation	iv
List of Figure	v
List of Table	vi
List of Symbols	vii
1. Chapter - 1	8
1.1 Introduction	
1.2. Literature Survey	
1.3. Problem Definition	
1.4. Objective	
1.5 Proposed Solution	
1.6 Technology/Platform used	
2. Chapter - 2	11
2.1 System Design/Block diagram	
2.2 Flow Chart	
2.3 Software Requirements	
2.4 Cost Estimation	
3. Chapter - 3	14
3.1 Implementation Snapshots/figures with Explaintion	
3.2 Implementation Code	
3.3 Results and Discussion	
3.4. Future Direction	
4. Chapter - 4	19
4.1. Conclusion	
Reference	20

List of Figures

Sr. No.	Name of figure	Page No.
3.3.1	Block Diagram	xvi
3.3.2	Flow Chart	xvii
3.3. 3	S equence Diagram	xviii
3 .3.4	Data Flow Diagram (DFD)	xix
3.3 .5	Use-Case Diagram	xx

List of Tables

Sr. No.	Name of table	Page No.
1.1	Literature Survey	xxi

ABSTRACT

Traditional cloud-based file storage systems rely on centralized servers, making them vulnerable to security breaches, data loss, and privacy violations. This project presents a decentralized file storage and sharing system using blockchain technology and the InterPlanetary File System (IPFS). Files are encrypted, divided into fragments, and distributed across multiple peer-to-peer nodes to eliminate single points of failure. Blockchain smart contracts are used to manage access control and verify data integrity. The system ensures user data ownership, transparency, and immutability without reliance on third-party providers. Wallet-based authentication enables secure user access. By removing intermediaries, the solution reduces storage costs and enhances scalability. The proposed system also offers censorship resistance and high data availability. Overall, the project demonstrates a secure, cost-effective, and reliable alternative to traditional centralized storage solutions.

Chapter - 1

1.1 Introduction

In the digital era, the rapid growth of data generation has increased the dependence on cloud-based file storage systems for storing, sharing, and managing information. Traditional cloud storage platforms rely on centralized server infrastructures controlled by third-party service providers. While these systems offer convenience and scalability, they introduce several critical challenges such as security vulnerabilities, privacy concerns, high operational costs, and lack of user control over stored data. Centralized architectures also create single points of failure, making them susceptible to cyberattacks, data breaches, server downtime, and censorship.

To address these limitations, decentralized file storage has emerged as a promising alternative. Decentralized storage systems distribute data across multiple independent nodes rather than relying on a single centralized server. This approach enhances fault tolerance, improves data availability, and eliminates dependency on intermediaries. However, decentralization alone is insufficient without a secure and transparent mechanism to manage access, integrity, and trust among participants.

Blockchain technology provides a robust foundation for building secure decentralized systems. By utilizing cryptographic hashing, distributed ledgers, and smart contracts, blockchain ensures data immutability, transparency, and tamper resistance. When integrated with decentralized storage protocols such as the InterPlanetary File System (IPFS), blockchain enables content-based addressing, where files are retrieved using cryptographic hashes instead of physical storage locations. This significantly improves data integrity and prevents unauthorized data modification.

The proposed decentralized file storage and sharing system combines blockchain and IPFS to create a secure, scalable, and cost-effective storage solution. Files are encrypted before storage, split into multiple fragments, and distributed across peer-to-peer nodes. Smart contracts deployed on the Ethereum blockchain manage user authentication, access permissions, and verification of file integrity. Wallet-based authentication ensures secure and user-controlled access to data without relying on centralized identity providers.

Overall, this system aims to enhance data security, preserve user privacy, reduce storage costs, and provide censorship-resistant file access. By eliminating centralized control and empowering users with full data ownership, the proposed solution offers a reliable and future-ready alternative to conventional cloud storage systems.

1.1 Literature Survey

Sr. No	Title	Methodology	Advantages	Challenges
1	IPFS-Based Secure File Storage	Used IPFS with cryptographic hashing for decentralized file storage and retrieval	High data availability improved integrity, no single point of failure	Data persistence depends on node availability
2	Blockchain for Data Integrity	Implemented blockchain smart contracts to verify and control data access	Tamper-proof storage, transparent access control	High transaction cost and scalability issues
3	Cost-Effectiveness of Decentralized Storage	Compared centralized cloud storage with decentralized storage platforms	Reduced long-term storage cost, better scalability	Performance depends on
4	Security Vulnerabilities in Centralized Storage	Analyzed security flaws in centralized cloud infrastructures	Identified key risks and improvement areas	Centralized systems remain prone to attacks
5	Censorship Resistance in Decentralized Networks	Studied blockchain-based decentralized networks	Prevents censorship and unauthorized data removal	Regulatory and legal challenges
6	Enhancing Data Security Using Decentralized Cloud Storage	Used encryption and redundant distributed storage	Improved reliability and security	Increased complexity in

1.3. Problem Statement

Traditional cloud-based file storage systems rely on centralized servers, which expose user data to security threats such as hacking, data breaches, and unauthorized access. These systems limit user control over stored data, impose high storage costs, and enforce access restrictions or censorship. Centralized architectures also suffer from single points of failure, leading to data unavailability during server outages. Hence, there is a need for a secure, scalable, and user-controlled file storage solution that eliminates dependency on centralized service providers.

Objectives.

1.4. Objectives

- To design a decentralized file storage and sharing system using blockchain technology.
- To ensure data confidentiality through encryption before storage.
- To provide secure access control using blockchain smart contracts.
- To eliminate single points of failure by distributing data across multiple nodes.
- To reduce storage costs while maintaining scalability and availability.
- To ensure data integrity, transparency, and censorship resistance.

1.5. Proposed Solution

The proposed solution is a decentralized file storage and sharing system that leverages blockchain technology and the InterPlanetary File System (IPFS) to overcome the limitations of traditional centralized storage. In this system, files are encrypted to ensure data confidentiality and then divided into multiple fragments before storage. These encrypted fragments are distributed across peer-to-peer nodes using IPFS, eliminating single points of failure.

Blockchain smart contracts are used to manage user authentication, access control, and file integrity verification. Instead of storing actual files on the blockchain, only file metadata and IPFS content identifiers (CIDs) are recorded, ensuring transparency and tamper resistance while reducing storage costs. Wallet-based authentication enables secure and user-controlled access to files.

Overall, the proposed solution provides enhanced security, data ownership, scalability, cost efficiency, and censorship resistance, making it a reliable alternative to conventional cloud storage systems.

1.6 Technology/platform used

- **Ethereum Blockchain:** Used for deploying smart contracts to manage access control, authentication, and file integrity.
- **Smart Contracts (Solidity):** Handle secure permission management and verification of stored files.
- **InterPlanetary File System (IPFS):** Provides decentralized, peer-to-peer storage using content-based addressing.
- **MetaMask Wallet:** Enables secure wallet-based user authentication and blockchain interaction.
- **Node.js:** Used for backend services and blockchain–IPFS integration.
- **HTML, CSS, JavaScript:** Used to develop the user interface for file upload and retrieval.
- **Cryptographic Techniques (AES, Hashing):** Ensure data confidentiality and integrity before storage.

Chapter - 2

2.1 System Design/Block diagram

The system design defines the architecture and workflow of the Decentralized File Storage and Sharing System using blockchain technology and IPFS. It explains how different system components interact to securely store, retrieve, and share files in a decentralized environment. The design emphasizes security, scalability, data integrity, and user control.

Key Components:

1. User Interface Module

- Provides a web-based interface for users to upload, download, and manage files.
- Enables wallet-based authentication using MetaMask.

2. Authentication & Access Control Module

- Verifies user identity through blockchain wallets.
- Manages file access permissions using smart contracts.

3. Encryption & File Processing Module

- Encrypts files before storage to ensure data confidentiality.
- Splits encrypted files into multiple fragments for secure distribution.

4. Decentralized Storage Module (IPFS)

- Stores encrypted file fragments across peer-to-peer nodes.
- Generates a unique Content Identifier (CID) for each file.

5. Blockchain Module (Ethereum Smart Contracts)

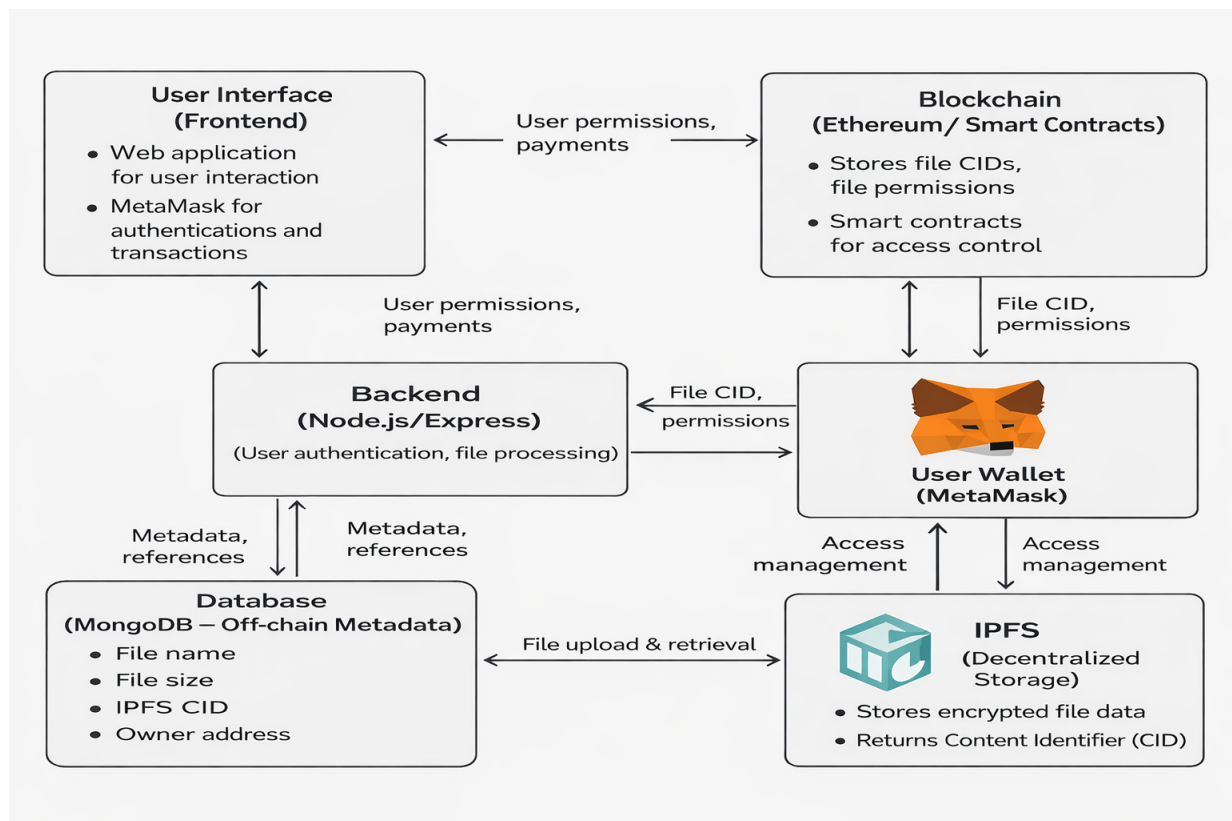
- Stores file metadata and IPFS CIDs on the blockchain.
- Ensures data integrity, transparency, and tamper resistance.

6. Retrieval & Decryption Module

- Fetches file fragments from IPFS using stored CIDs.
- Reassembles and decrypts files for authorized users.

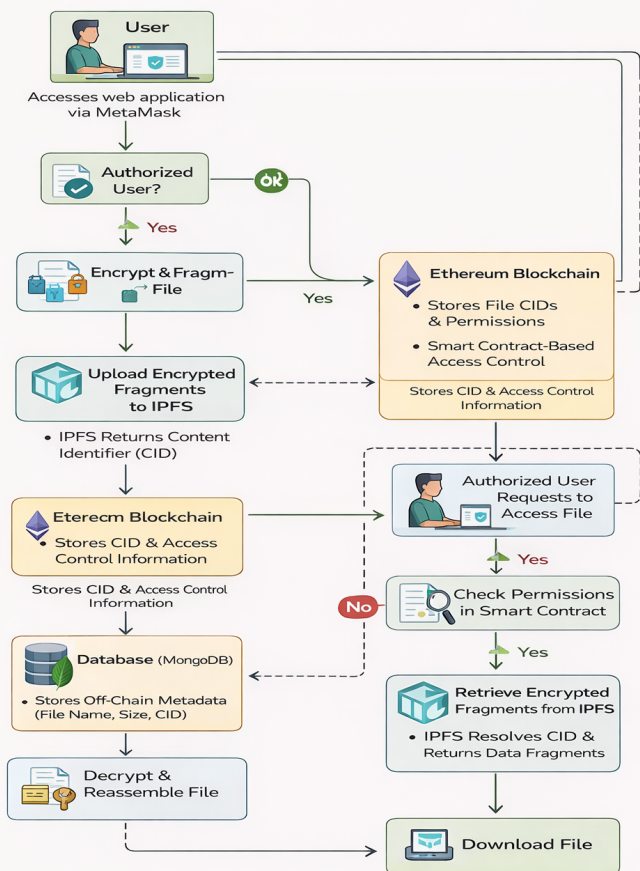
System Workflow:

1. User logs in using a blockchain wallet and selects a file for upload.
2. The file is encrypted and divided into multiple fragments.
3. Encrypted fragments are uploaded to IPFS, generating a unique CID.
4. The CID and access permissions are stored on the blockchain via smart contracts.
5. When requested, authorized users retrieve fragments from IPFS.
6. The system decrypts and reconstructs the original file for secure access.

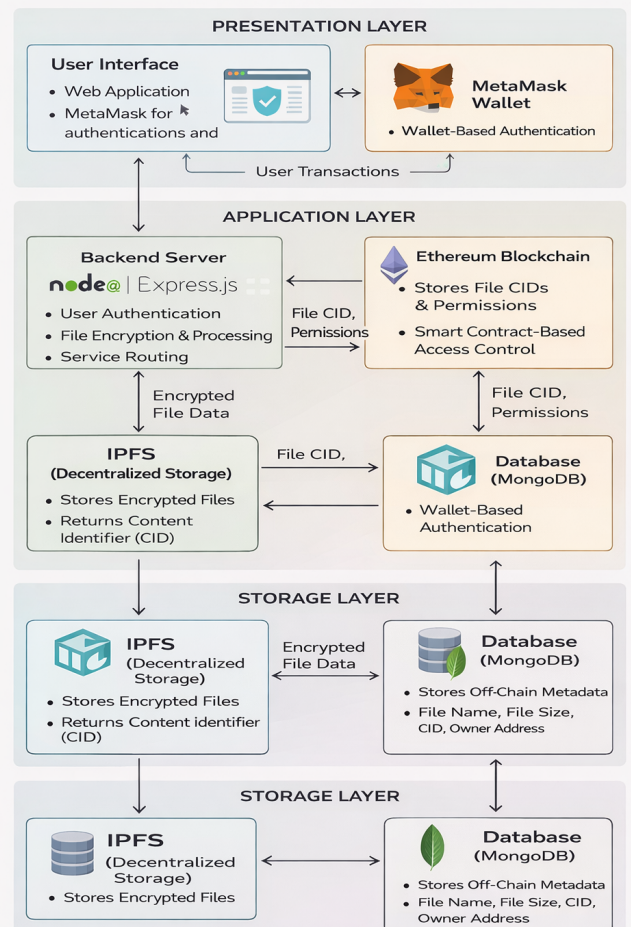


Block diagram

Flowchart of Decentralized File Storage & Sharing System



System Architecture



2.3. Software Requirements

- **Operating System:** Windows 10 / Linux / macOS
- **Blockchain Platform:** Ethereum
- **Decentralized Storage:** IPFS
- **Smart Contract Language:** Solidity
- **Blockchain Wallet:** MetaMask
- **Development Framework:** Truffle / Hardhat
- **Web Technologies:** HTML, CSS, JavaScript
- **Backend:** Node.js
- **Browser:** Google Chrome / Firefox

2.4 Cost Estimation

Development Costs:

Smart contract development and blockchain integration.
UI/UX design and full-stack application development.

Storage Costs:

Centralized cloud storage involves higher costs due to infrastructure maintenance.
Decentralized storage offers lower costs with efficient scalability.

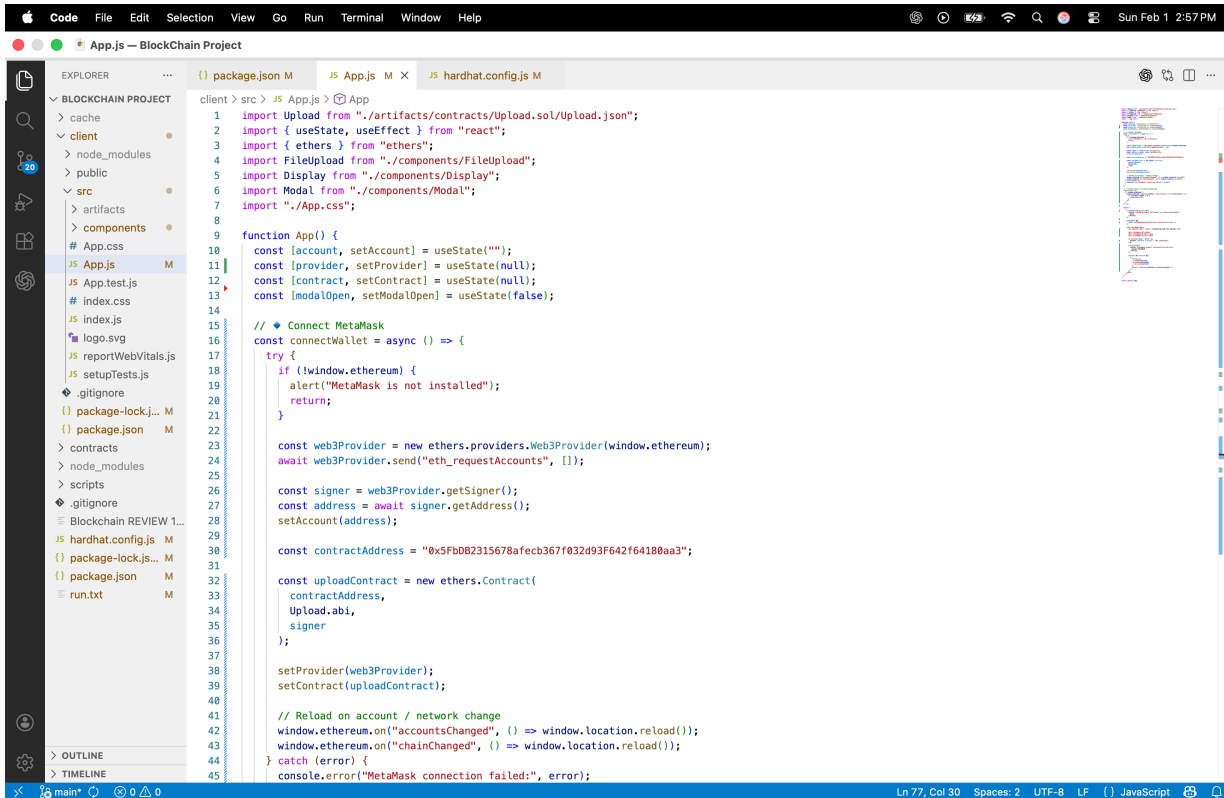
Transaction Fees:

Ethereum gas fees or the use of alternative Layer-2 scaling solutions.

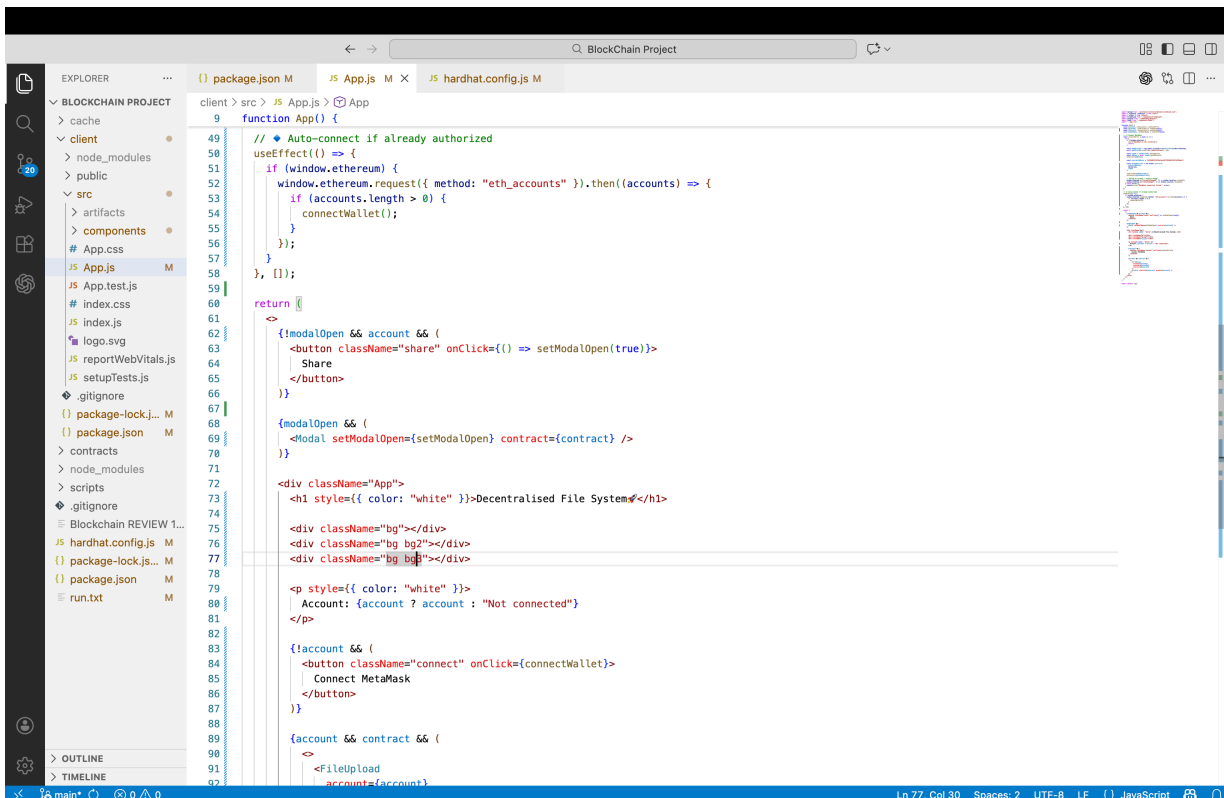
Hardware & Infrastructure:

Computing resources required for storage nodes and blockchain validation.

3.1 Implementation Code



```
1 import Upload from "../artifacts/contracts/Upload.sol/Upload.json";
2 import { useState, useEffect } from "react";
3 import { ethers } from "ethers";
4 import FileUpload from "../components/FileUpload";
5 import Display from "../components/Display";
6 import Modal from "../components/Modal";
7 import "../App.css";
8
9 function App() {
10   const [account, setAccount] = useState("");
11   const [provider, setProvider] = useState(null);
12   const [contract, setContract] = useState(null);
13   const [modalOpen, setModalOpen] = useState(false);
14
15   // Connect MetaMask
16   const connectWallet = async () => {
17     try {
18       if (!window.ethereum) {
19         alert("MetaMask is not installed");
20         return;
21       }
22
23       const web3Provider = new ethers.providers.Web3Provider(window.ethereum);
24       await web3Provider.send("eth_requestAccounts", []);
25
26       const signer = web3Provider.getSigner();
27       const address = await signer.getAddress();
28       setAccount(address);
29
30       const contractAddress = "0x5Fb082315678afecb367f032d93f642f64180aa3";
31
32       const uploadContract = new ethers.Contract(
33         contractAddress,
34         Upload.abi,
35         signer
36       );
37
38       setProvider(web3Provider);
39       setContract(uploadContract);
40
41       // Reload on account / network change
42       window.ethereum.on("accountsChanged", () => window.location.reload());
43       window.ethereum.on("chainChanged", () => window.location.reload());
44     } catch (error) {
45       console.error("MetaMask connection failed:", error);
46     }
47   };
48 }
```



```
49 // Auto-connect if already authorized
50 useEffect(() => {
51   if (window.ethereum) {
52     window.ethereum.request({ method: "eth_accounts" }).then((accounts) => {
53       if (accounts.length > 0) {
54         connectWallet();
55       }
56     });
57   }
58 }, []);
59
60 return (
61   <div>
62     <div className="bg">
63       <button className="share" onClick={() => setModalOpen(true)}>
64         Share
65       </button>
66     </div>
67     <div className="modal">
68       <Modal setModalOpen={setModalOpen} contract={contract} />
69     </div>
70     <div className="App">
71       <h1 style={{ color: "white" }}>Decentralised File System</h1>
72       <div className="bg">
73         <div className="bg bg2">
74           <div className="bg bg3">
75             <p style={{ color: "white" }}>
76               Account: {account ? account : "Not connected"}
77             </p>
78             {account && (
79               <button className="connect" onClick={connectWallet}>
80                 Connect MetaMask
81               </button>
82             )}
83             {account && contract && (
84               <FileUpload
85                 account={account}
86               />
87             )}
88           </div>
89         </div>
90       </div>
91     </div>
92   </div>
93 );
```

4.2 Outputs

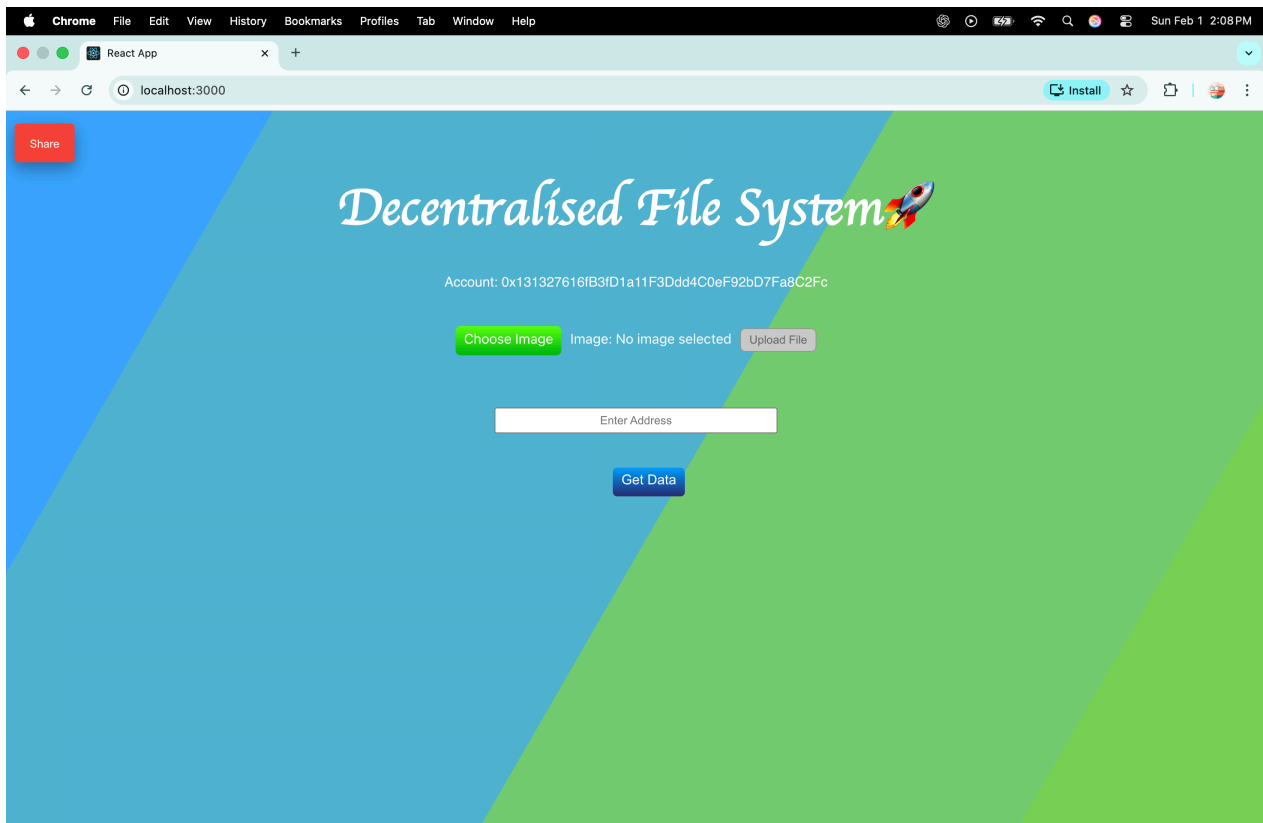
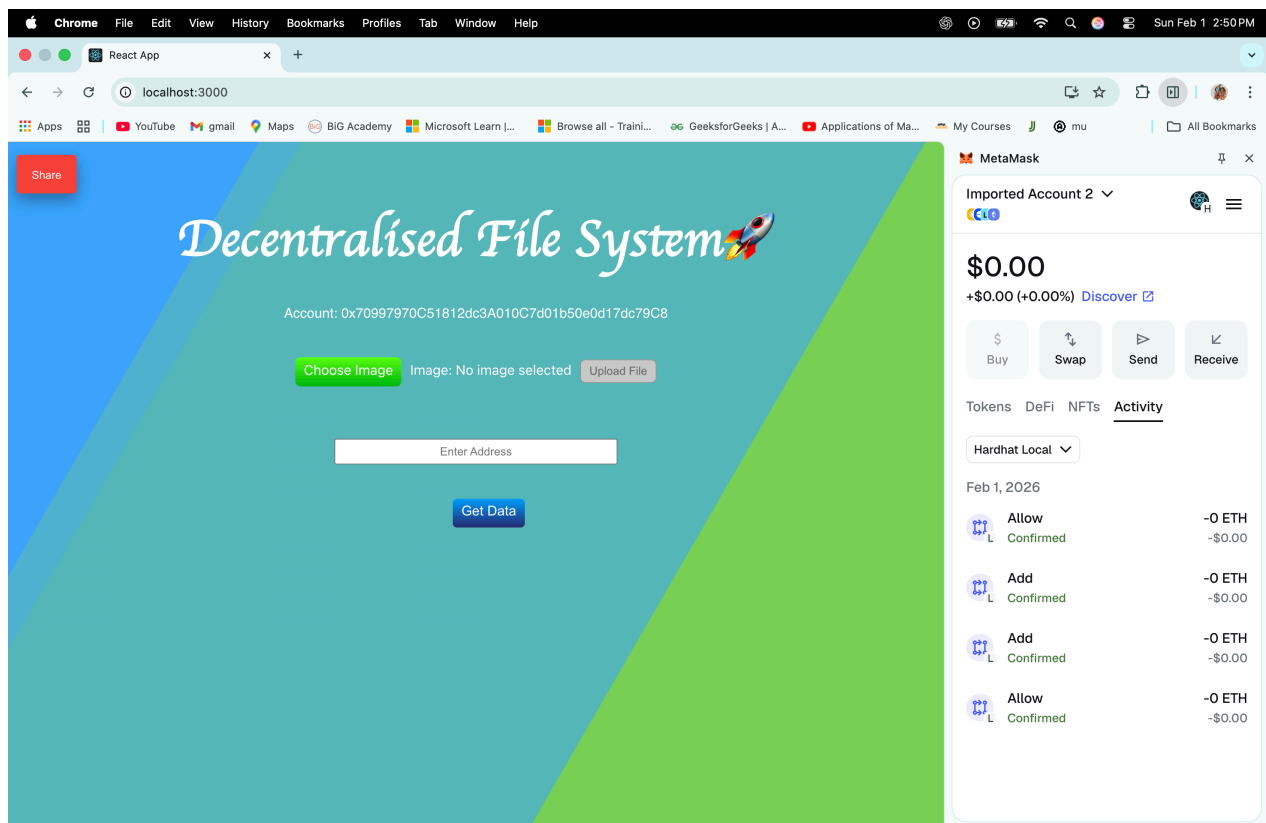
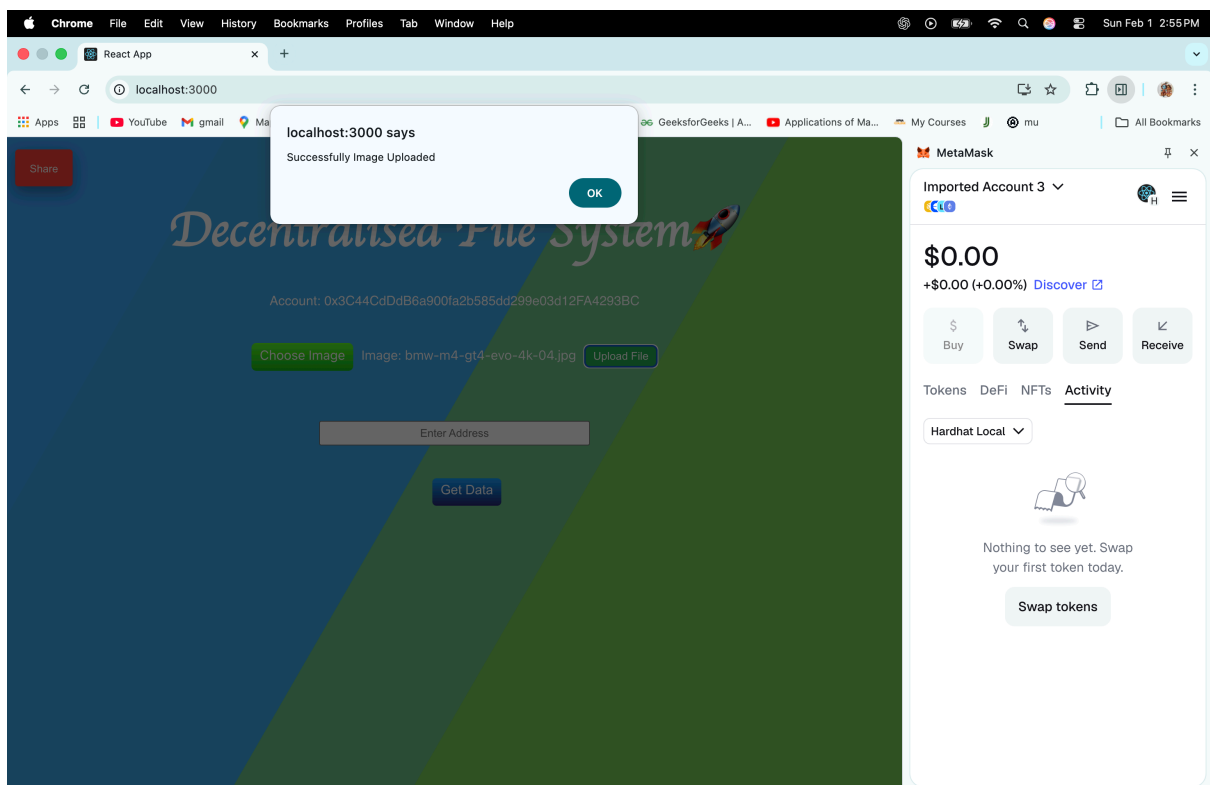
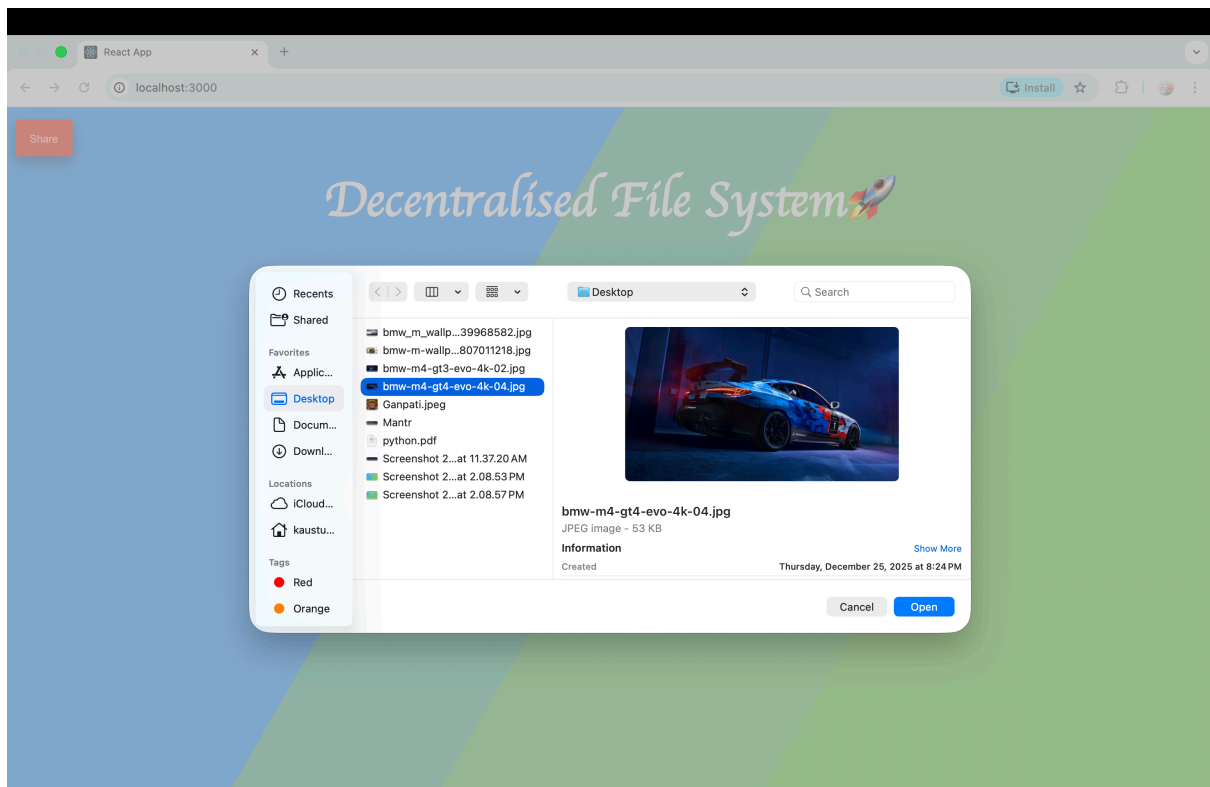
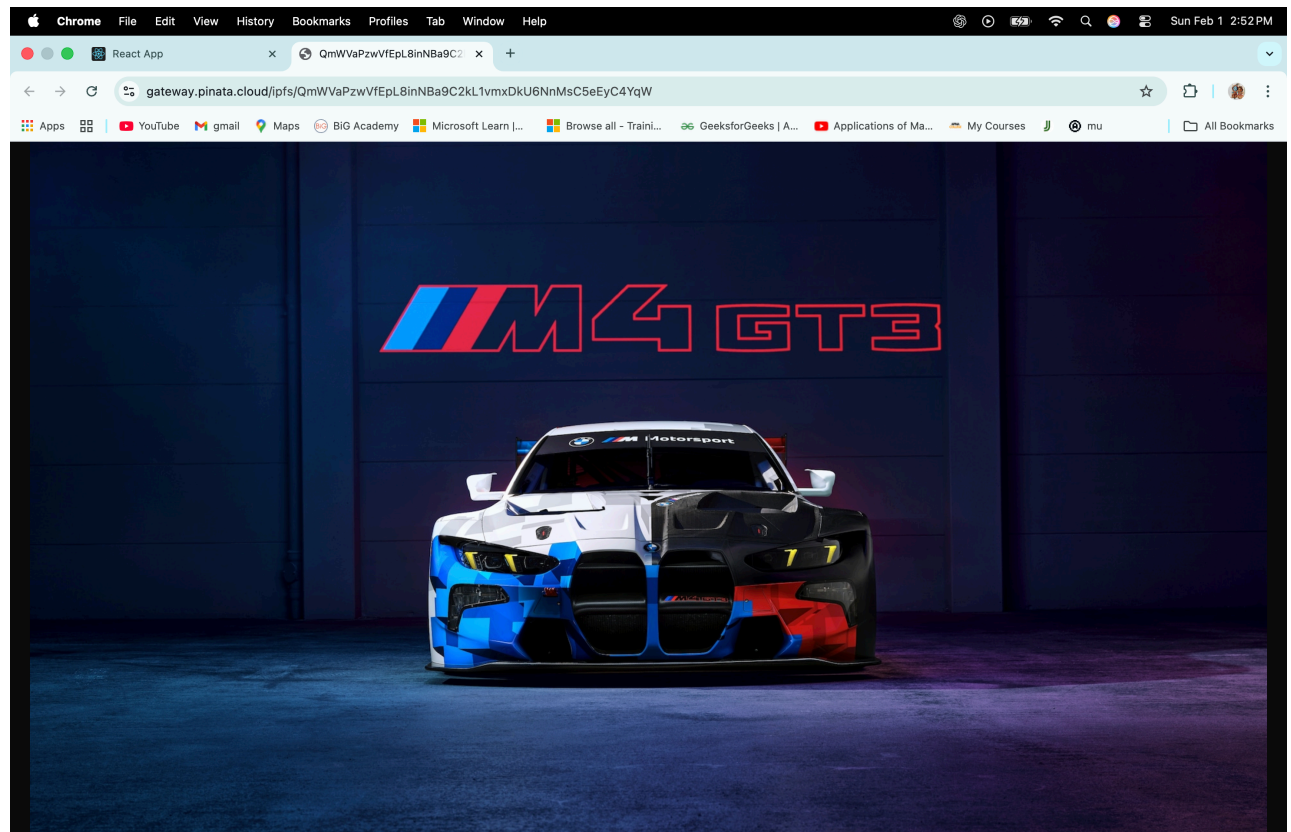
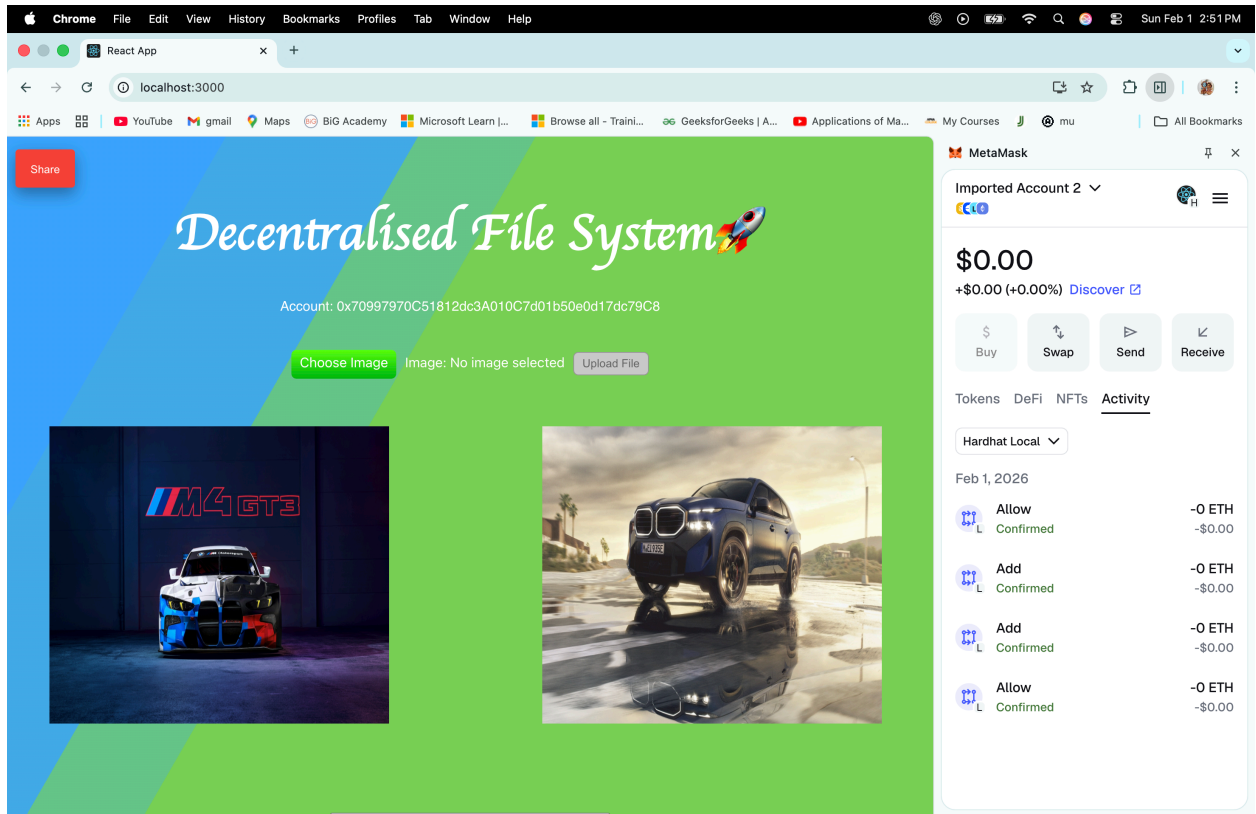


Figure 1: Home Page.







3.3. Results and Discussion

The proposed decentralized file storage and sharing system was successfully designed and implemented using blockchain technology and IPFS. The system demonstrated secure file upload, storage, and retrieval through wallet-based authentication and smart contract-controlled access management. File encryption and fragmentation ensured data confidentiality and reduced the risk of unauthorized access.

The use of IPFS enabled decentralized storage with content-based addressing, improving data availability and eliminating single points of failure. Blockchain smart contracts effectively managed file ownership, permissions, and integrity by storing Content Identifiers (CIDs) and access rules on the Ethereum network. Off-chain storage of metadata in MongoDB reduced blockchain storage overhead and transaction costs.

Experimental observations showed that decentralized storage provides better fault tolerance and data integrity compared to centralized cloud systems. However, transaction latency and gas fees were identified as performance constraints, particularly on the Ethereum main network. These limitations can be mitigated using test networks or Layer-2 scaling solutions. Overall, the results validate that the proposed system is secure, scalable, and cost-effective, making it a practical alternative to traditional centralized file storage solutions.

3.4. Future Direction

- Integration of **Filecoin or other incentive mechanisms** to reward storage providers.
- Use of **Layer-2 blockchain solutions** to reduce transaction fees and improve scalability.
- Implementation of **advanced encryption techniques** and automated key management.
- Development of a **mobile application** for easier file access and management.
- Support for **cross-chain interoperability** between different blockchain platforms.
- Addition of **AI-based storage optimization** for efficient data replication and retrieval.
- Enhancement of **user interface and user experience** for non-technical users.

Chapter - 4

4.1 Conclusion

The decentralized file storage and sharing system presented in this project demonstrates an effective alternative to traditional centralized cloud storage solutions. By integrating blockchain technology with the InterPlanetary File System (IPFS), the system ensures enhanced data security, privacy, and integrity. File encryption, distributed storage, and smart contract-based access control eliminate single points of failure and unauthorized access. The proposed system provides users with full ownership and control over their data while reducing storage costs and dependency on third-party providers. Additionally, the decentralized architecture offers high availability, transparency, and censorship resistance. Overall, the project highlights the potential of blockchain-based decentralized storage systems as a secure, scalable, and future-ready solution for data management.

Key Achievements of this Project Include:

- Secure file storage using encryption and decentralized distribution via IPFS.
- Effective access control and data integrity management using blockchain smart contracts.
- Elimination of single points of failure through peer-to-peer storage architecture.
- User-friendly interface with wallet-based authentication for secure access.
- Improved data availability, privacy, and censorship resistance compared to centralized systems.

REFERENCE

1. Secure and Scalable Dual Blockchain & IPFS Driven System

Paper: Secure and scalable dual blockchain and IPFS driven IoT-based data protection system

Authors: S. R. Mallick *et al.*

Year: 2025

Publisher: *Scientific Reports (Nature)*

Highlights: Integrates blockchain with IPFS for secure decentralized data storage in healthcare IoT applications, reducing single-point failures and improving resilience.

2. Decentralized Authentication & Secure Distributed File Storage

Paper: Decentralized Authentication and Secure Distributed File Storage for Data Integrity

Authors: M. Alsabaan *et al.*

Year: 2025

Publisher: *ScienceDirect (Computers & Security)*

Highlights: Proposes a blockchain-based IPFS system with enhanced access control and defense against poisoning and unauthorized access attacks.

3. Secure File Sharing Using Blockchain and IPFS

Paper: Secure File Sharing Using Blockchain and IPFS with Smart Contract-Based Access Control

Authors: **(Various)**

Year: 2025

Publisher: *Journal of Information Systems & eBusiness Management (JISEM)*

Highlights: Presents AES-256 encryption, SHA-256 hashing, and smart contract access control for data confidentiality and integrity.

4. Comparative Analysis of Decentralized Storage Platforms

Paper: Blockchain-Based Decentralized Storage Systems for Sustainable DSS

Authors: M. M. Merlec *et al.*

Year: 2024

Publisher: *Sustainability (MDPI)*

Highlights: Systematic comparative review of blockchain-enabled decentralized storage (IPFS, Filecoin, Storj, etc.) and their contributions to data sovereignty.

5. Integrated Blockchain-IPFS Framework for Privacy

Paper: Blockchain and InterPlanetary Framework for Secure & Privacy-Preserving Data Systems

Authors: S. Sayed *et al.*

Year: 2025

Publisher: *MDPI Journal of Security*

Highlights: Integrates group signatures with IPFS and blockchain to enhance privacy, authentication, and secure data sharing in sensitive applications.